

VLSI Devices

Lecture 23

Sung-Min Hong (smhong@gist.ac.kr)

Semiconductor Device Simulation Laboratory

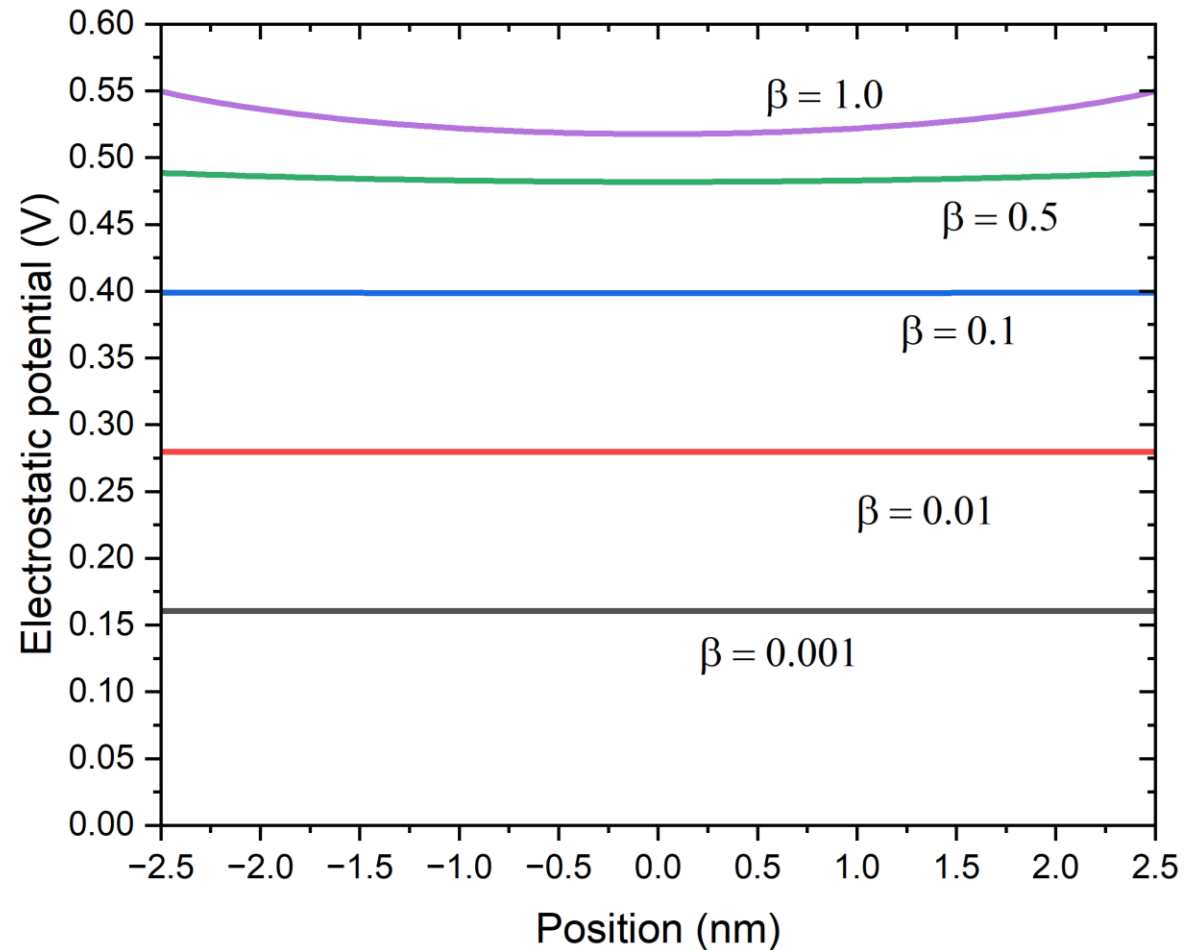
Department of Electrical Engineering and Computer Science

Gwangju Institute of Science and Technology (GIST)

L23

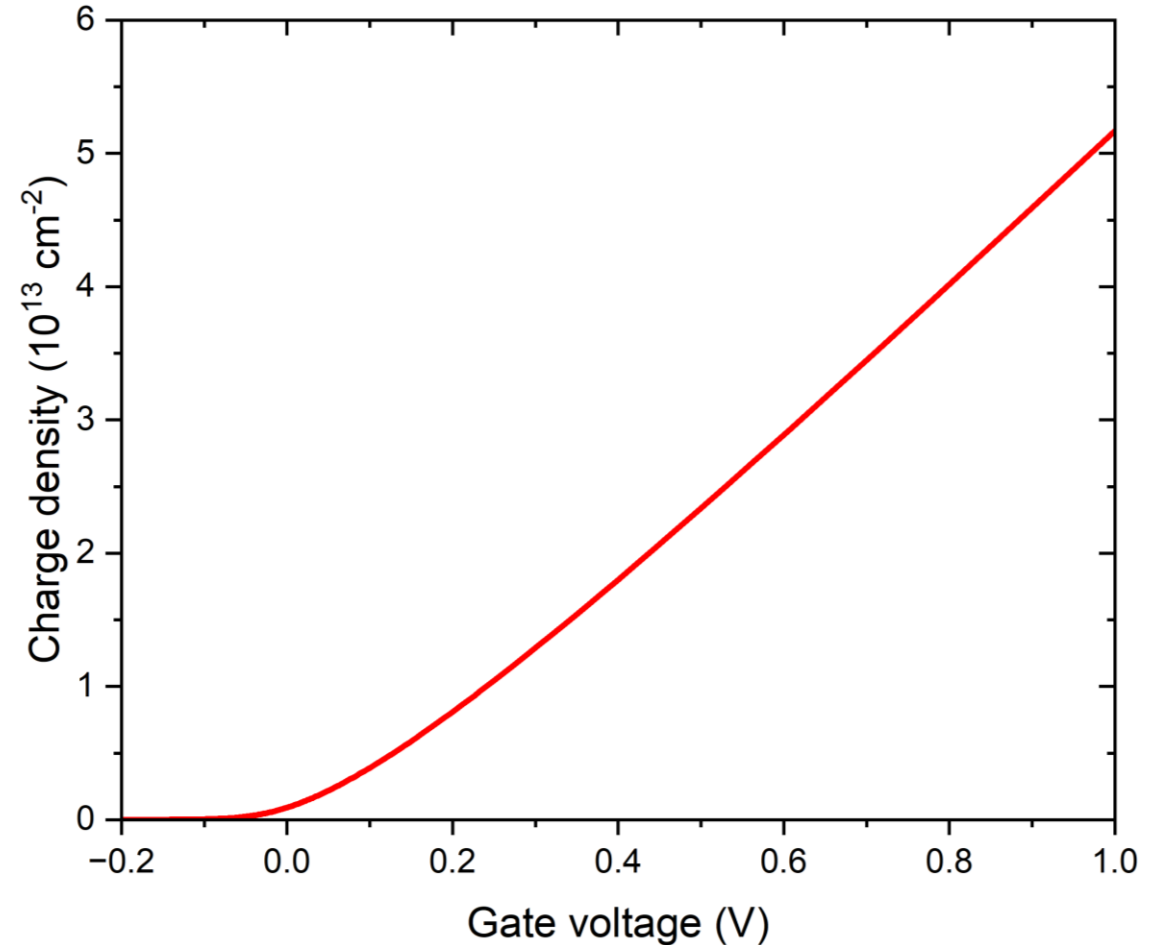
Answer of Homework#22 (1)

- $\phi(x)$ with various β values
 - Almost flat at low β



Answer of Homework#22 (2)

- Charge density
– β increases with V_{gs} .



Drain current

- Remember that

$$I_d = \mu_{eff} \frac{W}{L} \int_0^{V_{ds}} [-Q_i(V)] dV \quad \text{Taur, Eq. (5.10)}$$

- Using β ,

$$I_d = \mu_{eff} \frac{W}{L} \int_{\beta_s}^{\beta_d} [-Q_i(V)] \frac{dV}{d\beta} d\beta \quad \text{Taur, Eq. (7.23)}$$

- It is found that

$$-Q_i(V) = 2\epsilon_{si} \left. \frac{d\phi}{dx} \right|_{x=\frac{t_{si}}{2}} = 2\epsilon_{si} \frac{2k_B T}{q} \frac{2\beta}{t_{si}} \tan(\beta) \quad \text{Taur, Eq. (7.24)}$$

$$\frac{dV}{d\beta}?$$

- It is found that

$$-\frac{q}{2k_B T} \frac{dV}{d\beta} = \frac{1}{\beta} + \tan \beta + \frac{\epsilon_{si}}{\epsilon_{ox}} \frac{2t_{ox}}{t_{si}} \frac{d}{d\beta} (\beta \tan \beta)$$

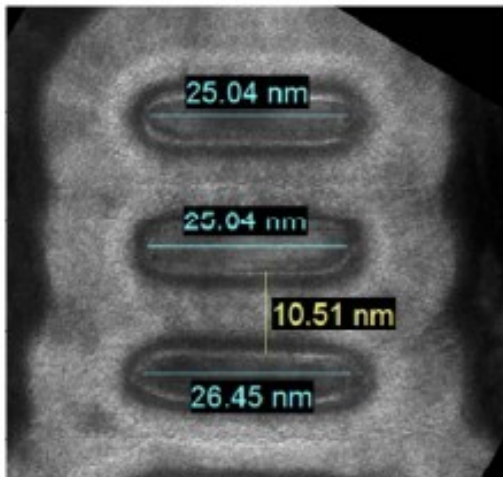
– Then,

$$\begin{aligned} I_d &= - \left(\frac{2k_B T}{q} \right)^2 \frac{4\epsilon_{si}}{t_{si}} \mu_{eff} \frac{W}{L} \int_{\beta_s}^{\beta_d} [\beta \tan \beta] \left(\frac{1}{\beta} + \tan \beta \right. \\ &\quad \left. + \frac{\epsilon_{si}}{\epsilon_{ox}} \frac{2t_{ox}}{t_{si}} \frac{d}{d\beta} (\beta \tan \beta) \right) d\beta \\ &= - \left(\frac{2k_B T}{q} \right)^2 \frac{4\epsilon_{si}}{t_{si}} \mu_{eff} \frac{W}{L} \left[\beta \tan \beta - \frac{\beta^2}{2} + \frac{\epsilon_{si}}{\epsilon_{ox}} \frac{t_{ox}}{t_{si}} \beta^2 \tan^2 \beta \right] \Bigg|_{\beta_s}^{\beta_d} \end{aligned}$$

Taur, Eq. (7.25)

Modeling of multigate MOS

- Various cross sections
 - Double-gate, nanowire, ...
 - Too many models!
 - Geometry does matter.
- Now, era of nanosheet MOSFETs
 - Cross-section varies.



Three layers of silicon
channel (Zhang,
IEDM 2017)

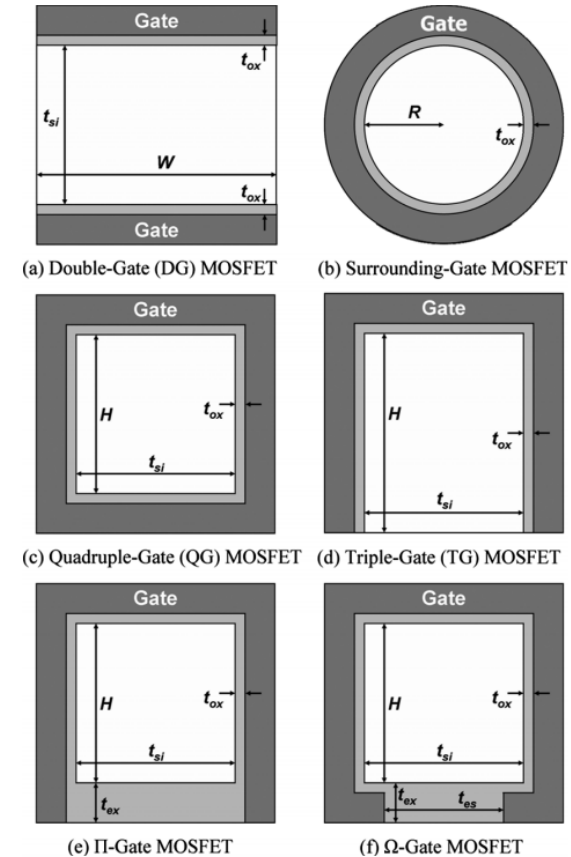
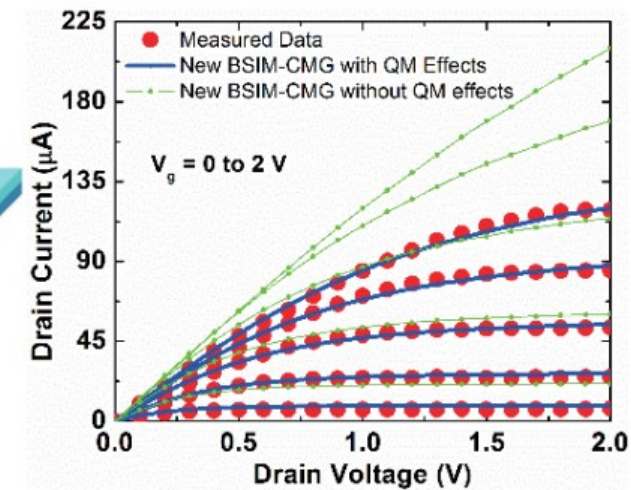
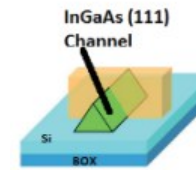
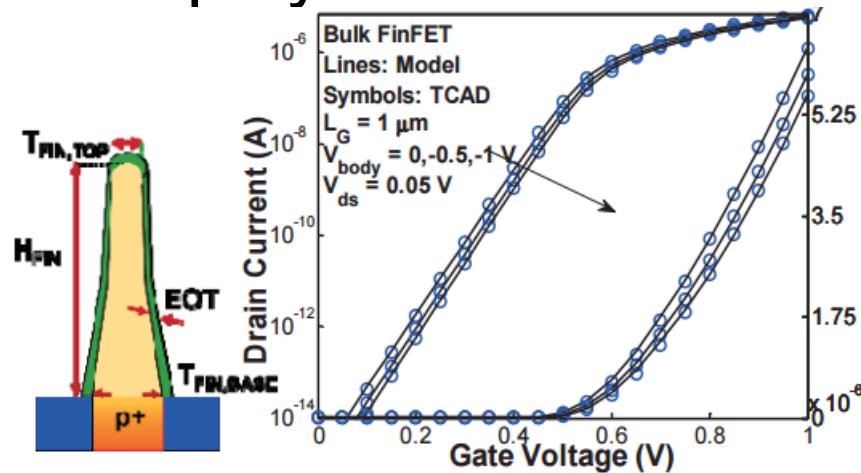


Fig. 13. Schematic diagrams of MG MOSFET cross-sections perpendicular to the channel direction.

Various cross sections
(Song, IEEE TCASI, 2009)

BSIM-CMG

- Standard compact model
 - It employs the “Unified FinFET Compact Model.”



Various devices tested using BSIM-CMG (VLSI 2015)

// Model Parameters for Unified FinFET Compact Model by Juan Duarte
10/2013

case (GEOMOD)

0: begin // Double Gate

if (!\$param_given(TFIN_TOP) || !\$param_given(TFIN_BASE)) begin

Weff_UFCM = 2.0 * HFIN;

Cins = Weff_UFCM * EPSROX * `EPS0 / EOT;

BSIM-CMG code →
(BSIM website)

GIST Lecture

Basic idea

- Unified approach based upon the “equivalent double-gate (DG) MOSFET” concept
 - Every multi-gate MOSFET is treated in the same way.
 - Parameters for the equivalent DG are extracted from the original cross section.
- Equivalent DG?
 - *A DG MOSFET whose electrical characteristics are the same with those of the original structure.*
 - Practically, it is known that the equivalent DG preserves:
 - Semiconductor area of the cross section
 - Perimeter of the semiconductor-insulator interface under the gate

Universal charge model

- Arbitrary cross-section

- It can be rigorously derived:

$$V_{gs} - \Phi_{mi} + \frac{Q_d}{P\langle C_{ins} \rangle_s} + \frac{Q_e}{P\langle C_{ins} \rangle_s} \approx V + V_T \log \left(\frac{\frac{1}{2} Q_e^2 + (1 - \alpha_e) Q_e Q_d}{q \epsilon P^2 V_T n_{int} \left(1 - \beta \exp \left(\frac{A^*}{P^2} \frac{2\alpha_e Q_e + Q_d}{2\epsilon V_T} \right) \right)} \right)$$

- A^* is the area, P is the perimeter, and $V_T = \frac{k_B T}{q}$

- α_e and β are fitting parameters.

Thank you!